PROJECT SUMMARY · 1 NOVEMBER 2026

What the pipeline misses

An audit of the standard de novo protein binder design pipeline, asking a single question: when the target surface is mutated, does any stage of the pipeline notice? The results, the structural weaknesses they expose, and what would have to change.

ONE

The protein, and the goal

IL-6 is a human signalling protein, one of the molecules the body uses to switch on inflammation. When it is stuck in the "on" position it drives rheumatoid arthritis and several other diseases, so it is a well-validated and well-characterised drug target.

The aim was to design a **small new protein from scratch** — around 50 residues, roughly a tenth the size of an antibody — that binds one specific patch on IL-6's surface, **site III**. Within that patch, **tryptophan 157** does most of the work: tryptophan is the largest of the twenty amino acids, a flat hydrophobic ring projecting from the surface. Any binder at this site has to engage that ring.

Nothing has been made in a laboratory. All of this is computational, and the subject of the audit is the software, not the molecules.

TWO

How the design pipeline works

The standard approach chains three programs, each doing one job and handing its output to the next.

1. Generate a shape

RFdiffusion invents the three-dimensional backbone of a new protein positioned against the target's surface. The output is a folded ribbon with no sequence attached.

2. Design a sequence

ProteinMPNN chooses which amino acid occupies each position along that backbone, so the shape will fold and hold together.

3. Filter

AlphaFold is given the designed pair and asked to predict its structure. If the prediction matches the design and the model is confident about the contact region, the design passes. Published work reports that this filter is the one correlating with experimental success.

That third step is the only gatekeeper in the whole process, and most of what follows concerns what it does and does not measure.

THREE

What was run

- **900 backbones** for the stage 1 experiment: three arms of 300, differing only in how the target residue was described.
- **47 backbones** carried forward, with around eight candidate sequences designed for each, giving approximately 380 designs.
- **319 complete designs scored** by AlphaFold in a single run — the entire pool rather than a selection.
- **27 further predictions** for the mutation experiment: nine designs that passed the filter, each in three conditions.
- **Six additional target proteins**, to establish whether conclusions drawn on IL-6 generalise or are specific to it.

Run on a university GPU cluster, with finished designs inspected structurally in a molecular viewer.

FOUR

The mutations, and the reasoning behind them

The question is straightforward. **If the target is damaged, does the software notice?**

Remove the residue a design was built to grip, and a method that genuinely evaluates binding should report that the design is now worse. A method that is really measuring "can I predict this structure confidently" may not respond at all. Distinguishing the two requires the experiment.

Three conditions were used:

- **Tryptophan 157 to alanine.** The large gripping residue replaced by the smallest one. This should matter a great deal.

- **Lysine 120 to alanine, as a negative control.** Chosen by measurement: this residue sits around 20 ångströms from the nearest atom of any design, well outside the contact region. It should produce no effect. If it does, the method is responding to perturbation rather than to chemistry, and the experiment cannot answer anything.
- **A separation experiment.** Keep the residue labelled as tryptophan while deleting the ring itself. The chemistry is gone and the identity remains. This separates a response to the residue's *chemistry* from a response to its *identity*.

The third condition proved to be the only one that distinguished the two possibilities, and it produced the most surprising result of the project.

FIVE

What each stage turned out to know

Stage 1 responds to residue identity, not to chemistry

Across 900 designs: deleting the entire tryptophan ring while keeping its identity changed the output distribution **not at all**. Keeping the atoms and changing the label to alanine changed it **completely**, with a large and statistically unambiguous effect.

So the backbone generator does respond to which residue occupies a position on the target — but it knows that residue as a categorical label rather than as an object with a size and a shape. It has learned, statistically, how binders tend to sit near "a tryptophan". It is not evaluating the ring.

Examining the source confirmed the mechanism. The program is given the target's full sequence as a categorical encoding. But the instruction that specifies *which residues to target* carries **one binary flag per residue and no chemical information at all**. It can be told "aim here"; it cannot be told "aim at something large and hydrophobic".

Stage 2 reads the chemistry, and is downstream of the decision

Changing that single target residue caused the sequence designer to rewrite roughly a fifth of the designed protein, on every backbone tested, with the random seed held fixed so the difference is attributable to the mutation alone.

The limitation is positional rather than representational. By the time stage 2 runs, the backbone is already committed. It can produce the best sequence for the shape it is given; it has no mechanism to reject that shape and request another.

Stage 3 does respond, and the negative control behaved

Nine designs that passed the filter, each scored in the wild-type condition, with tryptophan 157 removed, and with the distant control residue removed:

- Removing **tryptophan 157 degraded all nine designs**. Nine of nine in the same direction, with five showing effects an order of magnitude above the measurement noise.
- Removing the **distant control residue produced no systematic effect** — marginally worse in four designs and marginally better in five, consistent with noise.

The designs depend on the residue they were designed to engage, and the pipeline's final filter can detect it. This is the central positive result of the project.

The effect was not uniform, and the structure explains why. Designs on some backbones degraded severely; designs on one backbone barely responded. Measuring the interfaces directly, the unresponsive designs place far fewer atoms against the ring. They engage site III through a different binding mode. This is visible only because the spread was examined rather than the mean.

SIX

Loopholes in the pipeline

Each item below is a property of the tools, established by measurement, and each one will mislead a user who is not looking for it.

A co-folding filter could not distinguish a designed binder from a random peptide.

Tested as the scoring stage before AlphaFold: a **randomly generated 40-residue peptide scored higher than the actual design**, and the mutation chosen as an irrelevant control produced the largest effect of any mutation tested. Its interface confidence correlated strongly with its confidence in the whole structure. The method was measuring how hard the target is to fold, not whether the interface is real. Note that the on-target mutation alone would have read as a clean positive result; only the deliberately uninformative controls revealed otherwise.

The filter's scores are not reproducible.

Identical input, identical hardware, different answer. Most designs reproduce closely, but roughly **one in ten shifts by a large amount**, and one design moved far enough to cross the pass threshold without its structure changing. The distribution is heavy-tailed rather than well behaved, so a single design's score is not a verdict and any claim near the threshold requires replicate scoring.

The filter saturates, so it ranks poorly where ranking matters.

The metric is pinned near its ceiling for designs that are nowhere near the target site, and most variable in the narrow band where the pass threshold sits. It is a usable gate and an unreliable ranking, and the two are easily confused because they come from the same number.

Nothing verifies that the designed sequence folds back to the designed shape.

This check is not part of the pipeline and has to be added. Running it on our pool, **15% of sequences did not fold to the backbone they were designed for**. Those designs are indistinguishable from the rest at every earlier stage, and any conclusion drawn from one of them is drawn from a structure that does not exist.

A backbone cannot be evaluated from one sequence.

Within a single backbone, scores across eight sequences span a very wide range, so a pass belongs to the sequence rather than to the shape. Scoring the full pool demonstrated the cost of assuming otherwise: **the single most productive backbone in the set scored badly on its one test sequence, and four of its five sequences passed.** Screening shapes on one sample discards good shapes.

No stage computes an interaction energy, and adding one is not straightforward.

An explicit physics-based energy was added as an independent check. Established force fields **cannot score machine-learning-predicted structures directly**: the predicted geometry lies outside the domain the force field was parameterised for, and the resulting numbers are dominated by model strain rather than by the interface. The structure must be prepared specifically, and the comparison must be constructed symmetrically, before any value is meaningful. This is a real obstacle to bolting a physical energy onto a learned pipeline.

SEVEN

Why this pipeline is poorly suited to mutation questions

This is the conceptual finding, and it is independent of every individual weakness above.

> Nothing in the pipeline ever calculates the physical interaction between the two proteins. Not at any stage.

Considering what each stage does:

- Stage 1 generates a shape against a set of coordinates. It has no term for whether the two surfaces pack, attract or repel. It reproduces what binders statistically look like near particular residues.
- Stage 2 does see the target's chemistry, but only in order to write the best sequence for a backbone already committed. It cannot return the backbone.
- Stage 3 reports *confidence that it can predict the structure of the pair*. That is not evidence that they bind. A complex can be highly predictable and have no affinity.

The pipeline is therefore a shape generator, followed by a sequence designer operating on a fixed shape, followed by a predictability measure. At no point is complementarity between the two surfaces evaluated.

Two consequences follow, and both concern efficiency.

First, **success arrives stochastically and is detectable only at the end.** Of 319 designs, 16 passed. Within the backbone that produced the best designs, the sequences that engage the tryptophan well and those that do not were produced by the same program from the same shape, differing only in which random sample they came from. No earlier stage can separate them. The process does not steer; it samples and grades.

Second, and specifically for mutation work: **there is no stage that the question can be put to.** Establishing whether a design depends on a particular target residue requires building the design completely, mutating, re-scoring and comparing. The information becomes available only after all the expensive computation is finished. A pipeline that represented the target's chemistry at the point where decisions are made could answer far earlier, and could use the answer to produce a better design rather than merely to grade a finished one.

IN FAIRNESS TO THE PIPELINE

It works. Sixteen designs cleared a filter that published work reports as predictive of experimental success, the interfaces are of a realistic size, and the mutation experiment confirms they depend on the intended residue. The criticism is not that the pipeline fails; it is that it is **inefficient and uninformative**. It cannot explain why a design succeeded, cannot be steered towards success, and makes mutation questions far more expensive than they need to be.

EIGHT

What would have to change

- **Give the targeting instruction chemical content.** Specifying which residues to target currently carries one binary flag per residue. It could carry volume, hydrophobicity, charge and hydrogen-bonding capacity — the properties that determine whether two surfaces are complementary. This is a concrete and modest modification to an existing architecture, and it is the specific improvement this audit points to.
- **Introduce an explicit interaction energy before the final filter**, so that shapes can be rejected on physical grounds rather than graded on confidence after the fact.
- **Evaluate backbones on multiple sequences or not at all.** Scoring the full pool took under an hour, so there is no economy that justifies screening on a single sample.

NINE

Current status

- **16 candidate designs** through the calibrated filter, distributed across seven backbones, several strong on both the structural and the confidence measures.
- **The mutation question is answered.** The designs depend on tryptophan 157 and are unaffected by a distant control residue, measured across nine designs with a known noise floor.
- **The independent energy cross-check is in progress.** The method is validated; the run producing the numbers has not completed.
- **No experimental work has been done.** Every statement here concerns software behaviour, which is what the project set out to audit.

The full technical record, with every number, table and statistical test behind the claims above, is in IL-6 binder audit. An introduction to the project from the beginning is in Designing a protein that sticks.

Figures in this summary are rounded for readability; the technical record is authoritative.